

3.1.12 Acids and bases (A-level only)

Acids and bases are important in domestic, environmental and industrial contexts. Acidity in aqueous solutions is caused by hydrogen ions and a logarithmic scale, pH, has been devised to measure acidity. Buffer solutions, which can be made from partially neutralised weak acids, resist changes in pH and find many important industrial and biological applications.

3.1.12.1 Brønsted–Lowry acid–base equilibria in aqueous solution (A-level only)

Content	Opportunities for skills development
An acid is a proton donor. A base is a proton acceptor. Acid–base equilibria involve the transfer of protons.	

3.1.12.2 Definition and determination of pH (A-level only)

Content	Opportunities for skills development
The concentration of hydrogen ions in aqueous solution covers a very wide range. Therefore, a logarithmic scale, the pH scale, is used as a measure of hydrogen ion concentration. $\text{pH} = -\log_{10}[\text{H}^+]$ Students should be able to: • convert concentration of hydrogen ions into pH and vice versa • calculate the pH of a solution of a strong acid from its concentration.	MS 0.4 Students carry out pH calculations. MS 2.5 Students could be given concentration values and asked to calculate pH or vice versa.

3.1.12.3 The ionic product of water, K_w (A-level only)

Content	Opportunities for skills development
Water is slightly dissociated. K_w is derived from the equilibrium constant for this dissociation. $K_w = [\text{H}^+][\text{OH}^-]$ The value of K_w varies with temperature. Students should be able to: • use K_w to calculate the pH of a strong base from its concentration.	MS 0.1 Students use an appropriate number of decimal places in pH calculations. Students understand standard form when applied to areas such as (but not limited to) K_w MS 2.2 Students use $K_w = [\text{H}^+][\text{OH}^-]$ to find the pH of strong bases.

3.1.12.4 Weak acids and bases K_a for weak acids (A-level only)

Content	Opportunities for skills development
Weak acids and weak bases dissociate only slightly in aqueous solution. K_a is the dissociation constant for a weak acid. $pK_a = -\log_{10} K_a$ Students should be able to: - construct an expression for K_a - perform calculations relating the pH of a weak acid to the concentration of the acid and the dissociation constant, K_a - convert K_a into pK_a and vice versa.	MS 0.0 Students carry out pK_a calculations and give appropriate units. MS 0.1 Students understand standard form when applied to areas such as (but not limited to) K_a AT a, c, d, e, f and k PS 2.3 Students could calculate K_a of a weak acid by measuring the pH at half neutralisation.

3.1.12.5 pH curves, titrations and indicators (A-level only)

Content	Opportunities for skills development
Titration of acids with bases. Students should be able to: - perform calculations for these titrations based on experimental results. Typical pH curves for acid–base titrations in all combinations of weak and strong monoprotic acids and bases. Students should be able to: - sketch and explain the shapes of typical pH curves - use pH curves to select an appropriate indicator.	MS 3.2 AT a, c, d and k PS 3.2 and 4.1 Students could plot pH curves to show how pH changes during reactions.
Required practical 9 Investigate how pH changes when a weak acid reacts with a strong base and when a strong acid reacts with a weak base.	

3.1.12.6 Buffer action (A-level only)

Content	Opportunities for skills development
A buffer solution maintains an approximately constant pH, despite dilution or addition of small amounts of acid or base. Acidic buffer solutions contain a weak acid and the salt of that weak acid. Basic buffer solutions contain a weak base and the salt of that weak base. Applications of buffer solutions. Students should be able to: • explain qualitatively the action of acidic and basic buffers • calculate the pH of acidic buffer solutions.	AT a, c, e and k PS 1.1 Students could be asked to prepare and test a buffer solution with a specific pH value. MS 0.4 Students make appropriate mathematical approximations in buffer calculations.